

Daniel Yoo

647-981-3032 | jongsang.yoo@mail.utoronto.ca | [LinkedIn](#) | [GitHub](#)



Education

University of Toronto

Expected May 2028

B.A.Sc. in Mechanical Engineering (Mechatronics and Energy)

Toronto, ON

Relevant Coursework: Mechanical Design, Manufacturing Engineering, Analog and Digital Electronics for Mechatronics, Fundamentals of Computer Programming, Solid Mechanics

Technical Skills

Mechanical Design & Prototyping: SolidWorks (CSWA Certified), 3D printing, part & assembly design, engineering drawings, BOM generation, basic GD&T, DFM fundamentals, rapid prototyping, fixture and hardware setup development

Electronics & Hardware Validation: PCB design (Autodesk Eagle), oscilloscope-based testing and debugging, soldering, LTspice circuit simulation, analog signal validation, hardware prototyping and laboratory testing

Analysis & Computation: MATLAB, ANSYS Workbench (static structural & thermal FEA), engineering data analysis, validation and interpretation of results

Programming & Tools: Python (automation, data processing), Microsoft Excel (data organization, VBA automation), Microsoft PowerPoint

Professional: Ontario G Driver's License, Teamwork, Communication, Leadership

Technical Projects

Current Sensing Amplifier & Custom PCB (LTspice, Autodesk Eagle, Soldering) | [GitHub](#) Jan 2026 – Apr 2026

- * Designed an **LM324-based analog signal-conditioning circuit** that converts a 0–0.35 V sensor output into a 0.25–3.3 V microcontroller input range.
- * Designed and implemented an **active low-pass filter** (passband up to 1 kHz, ≥ 35 dB attenuation at 5 kHz).
- * Designed a **2-layer PCB** in Autodesk Eagle and generated **manufacturing-ready Gerber files** for fabrication.
- * Performed **soldering, PCB assembly, oscilloscope-based testing, and iterative debugging** to resolve signal instability and validate circuit performance.

PCB Pick-and-Place Machine (SolidWorks) | [GitHub](#)

Sep 2022 – Dec 2022

- * Designed a **600-mm desktop PCB Pick-and-Place Machine** integrating a rack-and-pinion Z-axis and belt-driven X–Y gantry system for precise component positioning.
- * Built a **complete SolidWorks mechanical assembly** (100+ components) and generated **manufacturing-ready engineering drawings and assembly documentation**.
- * Developed and validated **mechanical test setups** to evaluate positioning accuracy, repeatability, and subsystem integration performance.
- * Developed a **BOM** covering 60+ components across 5 subsystems and performed tolerance analysis to improve alignment and assembly reliability.

Parallel Gearbox Design and Assembly (SolidWorks, 3D Printing) | [GitHub](#)

Sep 2022 – Dec 2022

- * Designed a two-stage parallel-axis gearbox achieving a **16:1 speed reduction** using **dual 4:1 spur-gear stages** for controlled torque transmission.
- * Built a complete SolidWorks CAD assembly validating **gear meshing, shaft alignment, and motion transfer through integrated modeling**.
- * Applied **gear sizing and tolerance analysis** (± 0.2 mm) to ensure reliable rotation and mechanical performance in the fabricated prototype.
- * Prototyped structural components using **3D printing** and **optimized the housing design** to reduce part count by 25% while improving rigidity and assembly efficiency.

Experience

Engineering Workshop Lead

May 2025 – Present

Engineers Without Borders: UofT Chapter

Toronto, ON

- * Led 6+ engineering workshops for over 60 students, guiding **structured problem-solving and system-level design**
- * Developed and delivered **technical presentations and structured worksheets** to improve engagement and understanding
- * Managed **budget tracking and materials** using Excel, supporting multiple sessions with consistent cost control
- * Coordinated **team activities and facilitated collaboration** in group-based engineering challenges

Fire Direction Center (FDC) Specialist

Nov 2023 – May 2025

Republic of Korea Army — Artillery Unit

Gyeonggi, South Korea

- * Developed **Excel and VBA-based automation tools** to streamline artillery calculations, reducing manual processing time by approximately 40%
- * Managed and validated **100+ data entries per operation** to ensure accuracy in **time-critical decision-making**
- * Improved calculation accuracy and reduced human error through **structured data verification processes**
- * Integrated and processed multiple data inputs to support coordinated system-level operations